

Adjoint-Faithful FNO Project Tracker

MITGCM baroclinic double-gyre benchmark

Status date: 21 July 2026 Canonical plan: AF_FNO_Project_Plan.tex

Current position. The control, low-wind, and high-wind trajectory datasets (S0–S2) are complete. S0 reproduces the expected qualitative tutorial circulation and has passed all seven numerical validation gates. The immediate critical path is one shared preprocessing pass over S0–S2, followed by an adapted Bire-architecture baseline (A0). The intermediate-wind simulations I1–I2 and the response/adjoint programme follow only after this baseline is established. The local source tree has also been cleaned and prepared for its first remote Git commit.

1 Status legend

Status	Meaning
Complete	Finished and checked against the stated acceptance evidence.
Active	Workstream is open and may contain completed substeps.
Next	Immediate executable step on the critical path.
Pending	Planned, but intentionally not started.

2 Milestone dashboard

ID	Milestone	Status	Evidence / exit condition
M0	Reproducible MIT-GCM build and segmented runner	Complete	S0 build job 283675; four-rank executable and immutable per-segment provenance.
M1	S0 control trajectory	Complete	100-year spin-up plus 10-year daily production; jobs 283676–283686; final data inventory verified.
M2	Tutorial validation	Complete	Validation job 283958; seven of seven gates passed; four project-facing figures and machine-readable report.
M3	S1/S2 forcing regimes	Complete	Low- and high-wind adjustment/production jobs 283959–283962 complete; daily inventories verified.
R0	Git provenance baseline	Next	Clean first commit pushed to the project remote; generated data, upstream clones, builds, environments, and scheduler logs excluded.
M4	Shared ML dataset	Next	Convert, center C-grid variables, mask, split chronologically, and compute training-only normalizers for S0–S2.

ID	Milestone	Status	Evidence / exit condition
M5	A0 adapted Bire baseline	Next	Small-sample overfit test, then frozen paper-architecture baseline trained and evaluated on the shared dataset.
M6	A-C model ladder	Pending	Modern state-only model A, forward model B, then response-supervised AF-FNO model C.
M7	I1/I2 wind interpolation	Pending	Two unseen intermediate-wind trajectories generated and opened for inference only.
M8	Forward-response campaign	Pending	224 ten-day perturbed restarts: 128 training, 32 validation, and 64 inference.
M9	Blind adjoint evaluation	Pending	Frozen artifacts/checksums recorded before 12 true-adjoint tests are generated or inspected.
M10	Control-update verification	Pending	Three forward checks demonstrate whether emulator-gradient controls improve each objective.

3 Completed simulation record

3.1 S0: control wind, $\tau_0 = 0.100 \text{ N m}^{-2}$

Item	Recorded result
Execution	Build job 283675; spin-up jobs 283676–283685; production job 283686.
Model duration	100-year spin-up plus 10-year production: 110 model-years, using a 360-day year.
Production inventory	3,600 daily <code>dynState</code> records and 3,600 daily <code>surfState</code> records.
Measured size	4.6 GB in the raw MITGCM archive.
Validation	Job 283958; all seven tutorial gates passed.
Surface heat flux	Mean -1.9589 W m^{-2} .
Potential temperature	Final representative means: 17.6918°C at the surface, 8.3589°C at 305 m, and 2.4327°C in the abyss.
Gyre transport	First production-year extrema: $+32.679$ and -30.497 Sv .
Sea-surface height	First production-year range: -0.987 to $+0.762 \text{ m}$.

The temperature structure, double-gyre transport, sea-surface-height pattern, and late-spin-up trends are consistent with the intended official tutorial behaviour. The deep ocean remains the slowest-adjusting component, as expected; this is recorded as a scientific caveat rather than a failed gate.

3.2 S1 and S2: off-control wind regimes

Run	τ_0	Jobs	Schedule	Daily output	Size
S1	0.075 N m^{-2}	283959 / 283960	5-y adjust + 10-y production	3,600 dynState + 3,600 surfState	2.7 GB
S2	0.125 N m^{-2}	283961 / 283962	5-y adjust + 10-y production	3,600 dynState + 3,600 surfState	2.7 GB

Both branches restart from the validated S0 year-100 state. Their five-year adjustment periods are kept separate from the ten-year daily production records used by the ML pipeline.

3.3 Retired high-resolution Bire reconstruction

The earlier 0.25° Bire-oriented spin-up, formerly Slurm job **283514**, was cancelled by the project owner on 21 July 2026. Its scratch tree, compiled build, MITgcm input deck, scheduler wrappers, conversion/training wrappers, and generated reports were subsequently deleted. This campaign is closed and is not a prerequisite for A0 or AF-FNO. Only recovered source and documentation that provide architectural evidence for A0 are retained.

4 Data locations and provenance

Asset	Canonical location
S0 project entry point	outputs/af_fno/S0/
S0 result catalog	outputs/af_fno/S0/results_catalog.json
S0 validation	outputs/af_fno/S0/tutorial_validation/
S0 provenance copies	outputs/af_fno/S0/provenance/
S0 raw archive	/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S0/
S1 raw archive	/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S1/
S2 raw archive	/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S2/

The S0 project directory contains a symbolic link named `raw_mitgcm`; bulk output is not duplicated in `/home`. The catalog records counts, final pickup, hashes, executable provenance, and canonical paths. S1/S2 raw results currently reside under the corresponding scratch roots and should receive equivalent project-facing catalogs during preprocessing.

5 Frozen project decisions

- The active scientific benchmark is the official 1° , $62 \times 62 \times 15$ baroclinic double gyre. Exact recovery of the paper’s hidden 0.25° numbers is outside the primary claim; qualitative trends are the reproduction target.

- Model calendars use 360 days per year. The time step is 1,200 s, or 72 steps per model day.
- Standard trajectory production uses four MPI ranks. Long runs are split into restart-safe jobs that fit the two-hour **short** partition.
- S0–S2 share physics, grid, preprocessing, state definition, splits, and metrics. Only the prescribed forcing regime differs.
- True MITGCM adjoints are sealed test data. They are excluded from training, early stopping, architecture selection, hyperparameter tuning, perturbation-scale selection, and loss selection.
- All models must use the same immutable train/validation/test partition and training-only normalization statistics so architecture and loss comparisons remain interpretable.
- Git stores source, configuration, tests, small manifests, and project-facing documentation. Raw/reduced arrays, checkpoints, upstream clones, virtual environments, compiled trees, and scheduler logs remain external to Git.

6 New baseline decision: A0 before A–C

Decision added 21 July 2026. Establish an adapted Bire paper-architecture baseline, A0, before developing the modern NeuralOperator A–C sequence. This tests whether the published architectural idea recovers the paper’s qualitative wind and forecast trends on the available tutorial configuration.

A0 is a controlled adaptation, not a claim of exact historical reproduction. It keeps the recognizable Bire choices—channel-diagonal/separable spectral convolution, explicit pointwise channel mixing, three Fourier blocks, hidden width 128, positional encoding, and one-step state loss. Resolution-equivalent Fourier truncation uses 16 modes on the 62×62 grid. If coarse-to-fine pretraining is retained, the coarse grid uses stride 2 (31×31), rather than the paper code’s stride 8.

Changes forced by this benchmark must be disclosed: 46 Markov-state channels (centered U, V, temperature, and SSH), the 62×62 grid, ten production years per regime, wind stress as an input, and a ten-day primary comparison horizon. The obsolete public repository will be used as architectural evidence, not executed unchanged.

The comparison ladder is therefore:

Persistence / climatology $\rightarrow$ **A0: adapted Bire** $\rightarrow$ **A: modern state-only**
 $\rightarrow$ **B: forward losses** $\rightarrow$ **C: AF-FNO response supervision.**

Every transition changes a declared element, making gains attributable rather than confounding the dataset, architecture, and derivative-aware objective at once.

7 Immediate execution sequence

Current next step: shared S0–S2 preprocessing. No additional long MITGCM simulation is required before the first A0 training job.

1. Inventory and integrity-check every daily MDS record in the three production regimes; write equivalent S1/S2 result catalogs.
2. Center staggered U/V variables, apply wet masks, assemble the 46-channel state, and add forcing/geometry features required by the declared input contract.
3. Freeze chronological train/validation/test indices. Compute normalization from training indices only and record checksums for data, indices, and statistics.
4. Verify physical units, one-step tendencies, spectra, NaNs, masks, and restart continuity before exposing the dataset to a model.
5. Run an A0 overfit test on 20–100 samples. Require decreasing training loss and exact save/reload inference before scheduling full training.
6. Train and evaluate A0 against persistence and climatology; freeze its configuration and report before implementing A–C.
7. Generate I1/I2 and open them only for the predeclared unseen-wind interpolation evaluation.

8 Remaining simulation and evaluation programme

Work package	Count	Role
I1/I2 trajectories	2	Intermediate wind amplitudes; inference-only interpolation test.
Perturbed forward responses	224	Ten-day local responses: 128 training, 32 validation, 64 inference.
Mandatory true adjoints	12	Three smooth objectives $\times$ two horizons (10/30 days) $\times$ two unseen states.
Forward control checks	3	Test one emulator-gradient control update for each objective.
Optional true adjoints	6	Ninety-day extension only if mandatory criteria are met.

The minimum programme contains 244 scientific integrations in total: five trajectory runs (S0, S1, S2, I1, I2), 224 perturbed forward responses, 12 mandatory adjoints, and three forward control checks. S0–S2 account for three completed trajectory runs.

The mandatory adjoints alone are the ground-truth gradient dataset. They evaluate: an interior Gaussian SSH average, a smooth western-boundary SSH average, and an area-weighted barotropic gyre-strength objective. The emulator, evaluation code, pass criteria, and cryptographic hashes must be frozen before these results are opened.

9 Risks, corrections, and controls

Risk / issue		Control or recorded correction
Scratch is not archival		Raw S0–S2 data live on <code>/bigscratch</code> . Copy them to durable project storage before the cluster retention deadline; retain catalogs, hashes, manifests, and figures under the project tree.
Trajectory-year arithmetic		The proposal’s listed durations sum to 142 , not 132, model-years: $110 + 15 + 15 + 1 + 1 = 142$. Use 142 in compute/storage accounting.
Historical-code	incom- pleteness	Label A0 “adapted paper architecture” and record all forced changes; do not claim a bitwise or numerical reproduction of Bire et al.
Data leakage		Freeze chronological indices and derive normalization only from training years. Keep I1/I2 inference-only.
Adjoint leakage		Enforce the sealed-adjoint contract and checksum frozen model and evaluation artifacts before generating or viewing adjoint results.
Deep-ocean drift		Report depth-dependent trend diagnostics with forecast results; avoid implying full abyssal equilibrium from upper-ocean stationarity.
Large job count		Pack short response integrations into auditable Slurm arrays or bundles while preserving one manifest/result record per scientific integration.
Cross-HPC portability		UCSB absolute paths, modules, accounts, and partitions are not NASA-portable. Introduce site profiles, rebuild dependencies and executables on the destination system, transfer data separately, and pass smoke/restart tests before production.

10 How to update this tracker

Update this document after a milestone changes state, a scientific decision changes, or a material dataset/model artifact is created. For each update:

1. advance the status date and change only milestones supported by durable evidence;
2. record Slurm job IDs, model-time duration, output counts, storage location, and validation result for every completed simulation;
3. add artifact paths and checksums where available; never treat a scratch pathname alone as an archival record;
4. record decision changes in both the relevant section and the change log;
5. regenerate `Project_tracking.pdf` and check the LaTeX log for undefined references, overfull boxes, or missing files.

Operational details and evolving status belong here. Scientific definitions, acceptance criteria, and the intended experimental design belong in `AF_FNO_Project_Plan.tex`; keep the two documents consistent without duplicating extensive derivations.

11 Change log

Date	Type	Change
2026-07-21	Tracker created	Consolidated S0–S2 execution evidence, validation metrics, data locations, active decisions, remaining simulations, and risks.
2026-07-21	Method decision	Added A0, an adapted Bire paper-architecture baseline, before models A–C.
2026-07-21	Accounting correction	Corrected total trajectory duration from 132 to 142 model-years.
2026-07-21	Infrastructure	Prepared the source tree for Git: removed disposable Slurm/LaTeX/Python artifacts, strengthened ignore rules, replaced the paper-only README with the AF-FNO project overview, and documented NASA portability. The first remote commit remains R0.
2026-07-21	Campaign retired	Recorded cancellation of job 283514 and removed the obsolete 0.25° MITgcm scratch data, build, input deck, job wrappers, and reports. Retained only Bire-derived code evidence required for A0.